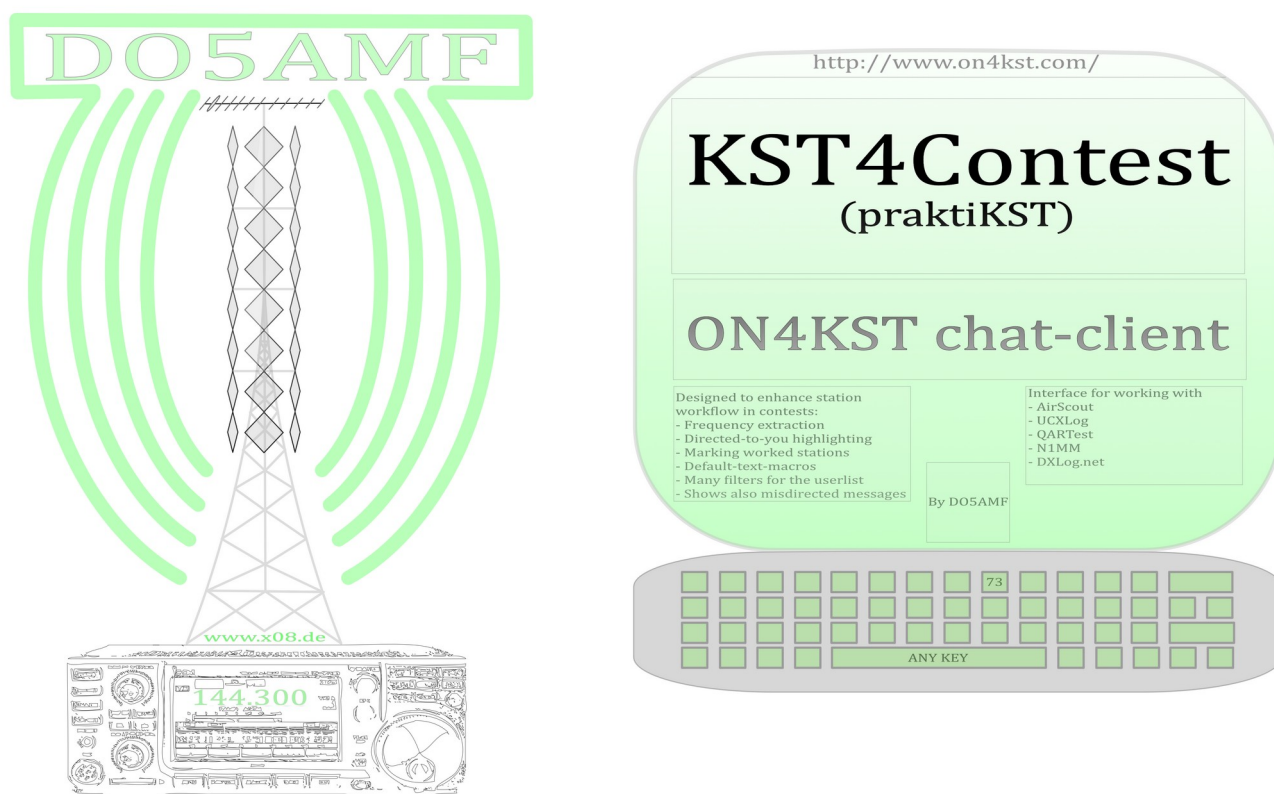

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1) Foreword

Historical note: this document records the development and knowledge available at GHz-Tagung Dorsten 2025. Ralf, DG6EA, noticed my work on the ON4KST chat and on the power supply at our contest station DM5M [1], and together we settled on the subject of an ON4KST chat client. This paper accompanies the presentation in Dorsten. It gives me room for rather more prose and a fuller explanation of where I came from and where I wanted to go. I have therefore written it as an extended version of the presentation. The title took some time to emerge. The original subject was the chat client itself, but neither the talk nor this paper is intended to be a product presentation or advertisement.[1]

It is nevertheless useful to explain the purpose of improving the contest workflow - and therefore the reason the program exists - by following the structure and functions of the software. Before doing so, I will describe the basis on which the implementation ideas rest. Some slide titles serve as headings in this paper.

I hope you enjoy reading it and attending what will be my first GHz-Tagung in Dorsten in 2025. 73 de DO5AMF, Marc

a) They called it “the chat”

My first contest activities, still at DF0GEB and initially as a spectator before I was licensed, date back to about 2003.

I mention this because I have taken part in VHF contests with DARC local group X08 almost continuously ever since, despite moving to Mannheim in 2005. Contesting continued as business as usual: for a long time with the DX Cluster at DB0NDH, and later via the Internet with DB0SUE.

In the good-faith belief that this was all the assistance available, we entered most VHF contests mainly for the pleasure of it and, as we see it, to operate our purchases and home-made equipment reliably over long periods. The latter has become increasingly important to me: after a 400 km journey, a contest should not become less enjoyable merely because something fails.

The 2024 season was particularly successful in that respect. We powered the complete station for 24 hours from about 1,750 Wp of solar panels and more than 15 kWh of LiFePO4 storage,

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including operation at the legal limit on VHF. Some of our generators are less resilient than the inverter now is. We hoped to continue this in 2025.

1) I was appalled

There is no better description. After DB0NDH and its packet-radio cluster were shut down, we were already expected to depend on the Internet. Mobile data was still expensive then and unattractive to most of us for this purpose alone. Now I learned that practically everyone except us traditionalists was using “the chat”. According to what I heard, many stations - particularly above 70 cm - relied on the Internet in order to make contacts. I did not think much of that.

Later I gave it a try and still did not particularly like it, although I certainly recognised the possibilities. Before I first saw a dedicated client, KST2Me, and knew only the web version, I was not impressed by the operating practices that chat use produced.

2) Some still are

Many years later, I still know plenty of radio amateurs who are appalled by the practice of basing amateur-radio operation on the Internet. Their passion often begins in the kHz range and ends, at the latest, near the upper edge of the 6 m band. Some were historically successful VHF testers and arranged skeds over HF, the next lower band or Packet Radio. They still expect the same of us today.

I know other stations that continue to call CQ for 24 hours on 23 cm using very long, narrow-beam antennas with opening angles below 15 degrees, without using the chat. Surprisingly, even this method somehow produces 15 to 20 QSOs in 24 hours.

Without being able to calculate the probability of a QSO, I would nevertheless assume that it is rather low.

The narrow antenna beams of two stations have to meet approximately, the QRG has to match, and one station must call CQ while the other searches for CQ - all at the same time.

The modern availability of transverters with spectrum displays is, of course, very helpful here.

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3) Chat: useful or necessary?

Among active operators on 144 MHz and above, acceptance of the chat is now higher. Mobile data is commonplace, and network coverage is adequate even in remote locations, although I have had to build a radio link to relay an LTE signal from a distant cell to our site.

Skeds are customary on 23 cm and above. Fortunately, the ON4KST chat has become the de facto standard platform. It can increase the number of QSOs per band through successful skeds on a previously unimaginable scale and, together with Aircraft Scatter, extend the workable distances just as dramatically. That is much more enjoyable than calling CQ for 16 hours to make one contact per hour. Even so, my enthusiasm is tempered by stations that, already on 144 MHz, no longer call or search for CQ with any ambition. I call this:

4) Active waiting

“Active waiting” is a term used by computer scientists. I use it for stations that rely entirely on being contacted in the chat, do not initiate messages themselves and then either:

- do not reply because they are not interested in the sked, for example because they want DX only;
- reply and complete the sked; or
- do not reply but nevertheless respond to the sked by answering the CQ call.

For such stations, this amounts to passive sked operation. In plain language:

If you do not contact these stations in the ON4KST chat, you will not make a QSO with them.

This does not surprise me on the higher and highest bands, nor in the milder form where a contest station does not call or search for CQ and only completes skeds while still seeking partners itself. CQ prospects are limited without arrays that have wide opening angles. On VHF the practice surprises me more, but it exists there too.

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5) Are repeated sked requests a nuisance?

It is therefore generally sensible to ask interesting unworked stations for skeds at reasonable intervals, even if they do not reply. Silence can have several causes:

- no interest in the sked;
- the station is not QRV and is merely reading the chat;
- the station does not reply in the chat but listens for your call;
- the station has a continuing pile-up and forgets to reply;
- the station overlooked the sked request, a subject to which I will return later.

There are surely others.

Holding back because one does not want to intrude or annoy others with requests is misplaced - within a reasonable interval, of course - because the requesting station does not know which operating mode the recipient is using.

Anyone logged into the chat during a contest must, can and generally wants to expect sked requests. Those who do not can mark themselves not QRV, identify the bands on which they are QRV or say that they want no further requests. KST2You even had an automatic reply for this, as I recall. Usually the unnecessary requests then stop.

6) Necessary versus laborious: the cost of using the chat

When we initially used the chat on 144 MHz only sporadically, I occasionally sat with DL5ZA at the 23 cm station in his caravan and acted as the chatter under DF0GEB. He operated and also chatted alongside. Because of UHF interference, he was separated from the rest of the team and did not object to an occasional visit and some help. I quickly understood why. Even the Microwave chat carried a great deal of traffic, and one had to read continuously to notice a message addressed to oneself.

I had not expected this mode of operation on 23 cm. I had imagined something relaxed. In reality, the total effort was greater than at our UHF station. At times so many messages arrived that looking away for 60 seconds risked missing that one had been contacted or mentioned in a reply. The phrase “reading slave” came to mind.

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At the time I could not quantify it. Today I can do so somewhat better. The following figures summarise four recent VHF contest chats. I saved the complete traffic and counted it; the figures are not exact to the last line.

July 2024, VHF and above

- 5 MB for 144/432 MHz
- 3 MB for Microwave
- more than 115,000 lines of text in total

September 2024 contest, VHF only

- 6.4 MB
- more than 96,000 lines

October 2024 contest, UHF and above

- 3.1 MB for 432 MHz
- 3.7 MB for Microwave
- more than 97,000 lines

Marconi Memorial Contest 2024, November, VHF CW

- 4.6 MB
- more than 67,000 lines

Respect to anyone who followed all of that with the web client.

In view of these figures, my subjective impression of a flood of messages is hardly surprising.

The number of messages is not the problem; the rate at which they arrive is. A sked takes considerable time when the client offers no good filters for extracting what matters to oneself. If the operator concentrates on radio operation and does not follow the chat, a message addressed to the station is easily missed.

Scrolling backwards to find the visual highlight then means moving through the past while new messages are missed. That is before considering misaddressed messages, to which I will return later.

Frequent screen changes tempt the operator to read everything. That can help when a nearby station works attractive DX that might also be possible from one's own location, but it leaves little time for one's own skeds. I will return to that too.

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The resulting operating practice becomes tiring and adds to the journey, setup, operation, dismantling and technical work. To quote DL5ZA approximately: “After a weekend like that, I always ask myself why I do it.”

The results of sked operation speak for themselves, however, and completing the prepared QSOs is rewarding. Operators who previously contested without the chat may nevertheless feel considerable reluctance to adopt this method.

There is also the staffing question: can and does a multi-operator station want to spare one operator per chat category, and is there someone eager to exercise a passion for reading operating abbreviations in rapid succession? In single-operator operation the problem is even more pronounced.

b) Operating practice and the chat

This brings me to “operating practice”. A compact definition might be: rules, procedures and sequences for proper radio operation.

The current examination question catalogue [2] asks: “What is the purpose of an amateur-radio competition, or contest? It serves competition and ... the continuous improvement of amateur-radio stations and operating practice.”[2]

End of story. The abbreviation “sked” does not occur in the questions. There are few questions in this area, which is understandable: preparation for the examination is only the entry to a many-sided hobby. The rest comes through operating.

In my view, arranging skeds - appointments for contest QSOs - is itself part of good operating practice. This also applies in multi-band operation, where a request can be made on a lower band on which contact with the partner already exists.

The objective of good VHF contest operating practice, limited by the available bands, is to maximise the total score when a good placing is the aim. In most subregional contests, the factors are kilometres covered and the number of QSOs per band, together with the bands activated because higher bands carry multipliers. In NAC contests, worked locator squares are another factor.

The operating practice and processes at a contest station can be improved. I will call the latter the workflow. The two ideas overlap, but operating practice includes more than station processes, such as language conventions, Q groups, abbreviations and instructions including “5 up”.

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c) Workflow at the station

A short definition would be a sequence of tasks used to achieve a particular objective. During a visit to DM5M in 2015, DJ5AR explained how and why improving his station processes had noticeably improved contest results, and that this was one of the areas offering the greatest gains.

A workflow develops through practice and keeps adapting. Sooner or later, for example, one learns to switch between VFO A and VFO B rather than tune across the dial when changing bands or QRGs. A further step in a multi-band contest may be to install more than one TRX so that a second band is available quickly or in parallel, or to omit “roger?” at the end of an exchange to save time. The workflow is optimised.

Software can also make the workflow at a contest station easier, support it and perhaps influence operating practice. Software can be developed with this objective in mind, which is what I am trying to do with KST4Contest.

d) Operating practice and workflow

Anything that serves an improved process at the contest station becomes part of its operating practice.

This begins with simple matters such as antenna direction. Suppose my antenna in JO51 points west. A station in SP calls and arrives 55 in SSB. I notice that the antenna is pointing the wrong way and do nothing.

A few years ago I would have turned east. Today, the improved process is to leave the antenna alone if the first contact works well. Our rotor would take about 40 seconds to turn through 180 degrees via its minimum, during which I would hear nothing. Without turning, the QSO is already complete.

The chat often has a more direct influence. The following example from the last Marconi Memorial Contest quotes two messages from 9A6A with comments.

- 2024-11-03 03:08:58Z 9A6A Petar (DM5M) pse loop
 - I needed his number. He could not hear what I needed.
 - He used the chat to instruct me to repeat my CW request continuously, without a time limit.
 - I sent “NR NR NR?” in a CW loop for about four minutes.
- 2024-11-03 03:14:02Z 9A6A Petar (DM5M) nw l loop nr
 - He had understood and sent his number as requested.
 - The QSB was severe, so he also repeated it continuously without a time limit.

Here the chat clearly influenced operating practice by providing instructions for my station, which I

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followed. At 1,060 km the path was too long for Aircraft Scatter, and the station heights above sea level - about 60 m at his end and 400 m at mine - were too low. We therefore had to use brute force to complete the QSO through the QSB. Naturally, no QSO data was exchanged directly through the chat. Even if this operating practice is already too Internet-heavy for some readers, it complies with the rules as long as the QSO data itself is exchanged exclusively by radio.

PS: My RST and serial number to Petar, and both QRA locators, were copied easily on the first pass. Conditions in MMC 2024 were unbelievably good.

Interim conclusion for operating practice: the chat should be watched so that changes in the counterpart's status are not missed. Instructions such as “hrd, pse ant to me” also offer considerable QSO potential because following them often makes it likely that a contact can begin.

e) Where can we improve?

First, the operator must decide whether the contest has an objective at all, and what it is. Only DX in the log? A high score? A good placing? As many bands as possible?

Many factors can be optimised for those objectives:

- Antenna setup

The antenna is the best amplifier, but often creates the greatest setup effort. DX needs high-gain antennas. Large QSO totals need many high-gain antennas covering several directions simultaneously, with switching, or broadly radiating antennas.

I chose the latter, specifically four-element quads [3], stacked in different numbers depending on the band. An opening angle of 69 degrees and about 12.0 dBi per antenna make them excellent when rapid setup, little space, acceptable gain and a wide beam are required. They are also extremely robust and need no matching network, so that cannot fail either.[3]

The combination of wide opening angle and gain creates receiving challenges, although even an IC-9700 handles them reasonably well today.

- Time budget

Does more always help? We have so far remained QRV for the full 24 hours, but not everyone sees it that way. A long operating time does not automatically produce a good placing or much DX. Some stations now enter six-hour classes. The question then becomes the best starting time. Statistically, I found that the first six hours of a 24-hour contest are among the most productive. Yet

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many stations cannot be reached then because they too enter the six-hour class and start at 07:00 UTC on Sunday.

- Power

More helps - well, not always. VHF and UHF also have low-power classes in which good placings are possible; we enter one on 70 cm. In our view, receive preamplifier performance is at least as important as transmitter power. A preamplifier should come before increasing equipment output, and we often run our PA well below the limit.

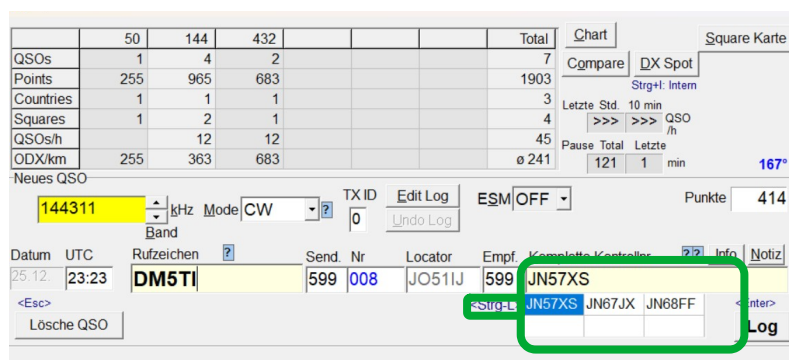
- Software and workflow

At last we reach the point. Considerable improvement is possible here without additional hardware. Logging software already plays a major role, and chat together with Aircraft Scatter can achieve much without great hardware expense.

After some experimentation, UCXLog [4] became our logger of choice. Most loggers have similar functions. We use UCXLog on VHF partly because its extensive station database includes locator history. Up to six additional locator suggestions appear for each callsign and can be toggled with Ctrl+L, as shown.[4]

When a locator can be recognised as the counterpart sends it, this offers a major advantage, particularly when heavy QRM makes copying uncertain.

Counterpart stations often operate repeatedly from the same location, so the locator may not need to be typed or toggled at all. This saves time, avoids input errors and improves the logging process and the rate at which stations can be worked.



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f) Review of available chat clients

- **ON4KST web client[5]**

The web client is a solid way to use the chat. Private messages are highlighted visually and acoustically. It is responsive and practical for everyday use, with several alarms for user login, locator login and text. During a contest with the traffic volumes described above, it demands a great deal of reading.

- **KST2Me by OZ2M[6]**

This solid desktop client is still used by many stations. I could test it only superficially because I do not own a licence. I would have bought one, but development and support stopped in 2021 and no new keys are issued.

- **KST2You by G1YFG[4]**

This web client is tidy and works on a smartphone or tablet; it was designed for that purpose and mainly addresses Microwave bands. It has QRB and private-message filters, a user list sortable by distance and a path-elevation diagram for the selected station or beacon. Technologically it is somewhat ahead of the ON4KST web client because it uses newer tools such as Bootstrap. It highlights the local callsign. For portable use on a phone or tablet, it is currently the best choice.

- **TUCNAK by OK1ZIA[7]**

TUCNAK is a good VHF contest logger with an integrated ON4KST chat client. This is practical and uses screen space very well. Private messages are highlighted, although they still disappear quickly during high activity.

- **KSTChat by US8AR[8]**

This is a very good desktop client in the style of a classic IRC client. It uses tabs, opening a new tab for each private conversation and filtering the text accordingly. The station's own messages are highlighted in the complete chat history, and an audible warning is provided.

The client also includes functions such as Aurora warnings and links to familiar VHF-and-above websites. It is well suited to everyday use. During contests, however, it requires considerable reading and creates much screen movement.

- **wtKST by DL8AAU and DL2ALF[9]**

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This is the oldest client in the list and has been maintained since 2012. It has several functions clearly intended for contest operation. Private messages are highlighted, filtered into a separate window and announced acoustically.

It is the only listed client with a mechanism for tagging logged stations in the user list so that worked stations are not contacted again. Another unique feature was its AirScout [10] connection, which indicates in the user list whether a station might be workable via Aircraft Scatter.[10]

Thanks to its link between the log and wtKST, this client was long my idea of the attainable optimum and to some extent the model for KST4Contest. I developed my own client only because wtKST was not compatible with UCXLog and manually marking logged stations was too cumbersome.

2) A short history of KST4Contest

We used wtKST regularly in contests from about 2018 whenever Internet access was available. Unfortunately, not every operator could be persuaded to mark worked stations consistently. As the station accumulated operating hours, it became harder to arrange skeds without receiving “qso b4”. This was a long-term problem I wanted to solve, although it was not the original purpose of my own client.

a) First iteration

The original objective was to extract more value from the chat itself. Instead of a complete chat client, I wrote a bare text client in my favourite language, Java, with functions for reading and analysis. It collected frequency information from stations to produce a private sort of DX Cluster.

I used regular expressions painstakingly assembled from historically recorded frequency formats.

It worked surprisingly well. The mechanism remains essentially the same, so readers can test how quickly frequencies accumulate. Even in early NAC tests with about 180 to 220 chat participants, most QRGs of active stations were usually collected within five minutes.

```
DL8QS, Heiko only 70cm in JO43KH @ QRG: 135
DL8SCQ, Ebi in JN48RV @ QRG: null
DL8UWE, Uwe 70cm in JO71DT @ QRG: null
DM5RC, Marcel in JO71ES @ QRG: null
DM5TM, Timon_70cm in JN59LE @ QRG: 220
DM7EE, Christian in JO52JJ @ QRG: null
DO1JKO, Juergen in JO52IK @ QRG: null
DO5AMF, Marc in JN49FK @ QRG: null
DR9A, 23/13/6/3 in JN48EQ @ QRG: 432.292
E70A, Darko in JN94FL @ QRG: null
E70AA, Zoran_70cm in JN93CQ @ QRG: 335
EA1IT, Ana in IN73TA @ QRG: null
EA2AGZ, Nicol. 2/70/23 in IN91DV @ QRG: null
EA2W, Jon in IN83XH @ QRG: null
EA3EZG, Jordi in JN01TP @ QRG: null
EI8KN, Roger in IO62IE @ QRG: null
EU1AI, Geo Minsk in KO33SV @ QRG: null
F1CBC, Philippe in JN09BO @ QRG: null
F1MKG, IARU 70/23 in JN08KQ @ QRG: null
F1NZC, Jean-Louis JN15 in JN15MR @ QRG: null
F1RDL, Pierre in JN25KU @ QRG: 1296.320
F1TRE, Philippe 70/23 in JN37PV @ QRG: null
F4BKV, Vincent in IN95PP @ QRG: null
F4CHA, 432/1296mhz IARU in JN08GS @ QRG: 1296.265
```

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As the example output shows, the resulting lists were rather unattractive from a usability point of view. That remained true even after I formatted and reduced the output. I enjoyed using it; nobody else gained much from it.

b) Second iteration

Nothing had yet changed the basic problem that arranging skeds became harder as a contest progressed. It was time to address it.

I still considered a graphical user interface a luxury, so the text client remained. It needed a mechanism to filter worked stations so they would not be contacted again and would disappear from the frequency lists.

wtKST provides an interface to Win-Test [11]. UCXLog has no logger API, so I examined its database files. A structure was visible but insufficient for a detailed interpretation without knowing the format. Worked callsigns, however, appeared as plain text.[11]

Having practised regular expressions for the frequency lists, I built one for complete callsigns with possible prefixes and suffixes. A file parser scanned the log database every 30 seconds, turned the results into a list and marked the callsigns in the chat user list as Worked. The function is still available.

This provided filtering, although with limits. It could not determine the band on which a callsign had been worked. From then on, the text client was used mainly at DM5M, not least for debugging.

The objectives were met, but the client was inconvenient. Monitoring took roughly as much effort as wtKST, with extra glances needed whenever a participant's QRG was to be captured.

That was not ideal and left room for workflow improvement. Better usability was needed to increase acceptance, including my own. A graphical interface had become unavoidable.

c) Third and subsequent iterations

Before adapting the software to our station workflow, I had to define where I came from and where I wanted to go, especially for the interface design but also for the functions behind it.

1) The five W questions

- Who should be able to use it? Should I publish it at all?

I had written the client solely for myself, mainly as an experiment. Other stations occasionally heard about the work, and questions became more frequent.

- What should it eventually do? Is a separate client worthwhile?

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When development began, I knew only the ON4KST web client and wtKST. As described, wtKST was close to what appeared possible. It was fair to ask whether reinventing the wheel made sense and, if so, which objective would keep the project from becoming pure entertainment.

- How should it be configured: XML file, database or user interface?

Making the project available to everyone implied a great deal of typing. The original configuration already used the XML file in which all settings are still stored. One cannot reasonably offer software that can be configured only by editing XML. Packet Radio may once have worked like that; today it no longer seems appropriate.

All functions needed to enter and save a station's operating parameters also had to be programmed. The amount of code is almost equal to the client functions themselves. Producing a user manual is similarly extensive.

- When should each feature be completed?

An objective matters if development is to progress, particularly when only one or two hours a day are available.

- Where should it be hosted? What about updates and release notes?

I had no web server at the time, so that too needed a solution. Fortunately, DO5ALF, my very first radio contact from CB days, was open to my ideas.

3) Principles behind the KST4Contest user interface

Once the rough direction was clear, work could begin. I chose the previously unfamiliar JavaFX [12] as the GUI framework. With platform independence as a requirement, Java and JavaFX seemed a good idea, and I rejected a browser application.[12]

Java applications have a good reputation for platform independence. Practice was sobering. At the time I had not managed to compile a Linux version or the frequently requested Windows 32-bit variant. I had invested little time in those issues because functions seemed more important.

The broad structure followed the ON4KST web client and wtKST, with adaptations. The following screenshots show how the design decisions were intended to improve the station workflow.

Let us begin with the main window:

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The screenshot displays the KST4Contest software interface. The top panel shows a message log with columns for Time, Callsign, Name, QRA, QRB, Message, Last known, and AP (minutes). The right panel shows a user list with columns for Callsign, Name, QRA, QTF, QRG, Act, AP (minutes / pot...), worked, and NOT QRV. The bottom left panel shows a snippets list with columns for Time, Callsign, Name, Message, and Last QRG. The bottom right panel shows user information for a selected station, including a list of messages and a status bar.

1. Private message

2. Snippets

3. User list

4. User information

Reading everything can be useful - for example if you are a computer.

All messages therefore pass through a message processor that extracts QRGs and can trigger other control functions.

For a person, reading everything becomes tiring. A fundamental design decision was therefore to display only what is relevant to me: my own messages and those addressed to everyone. The complete interface was designed to benefit the station's operating process.

a) Text templates and mouse obsession

This includes predefined text, because every contest repeats the same phrases, just as RTTY and CW do. I added text snippets from the beginning. Other clients call them macros. The standard snippets appear on the right.

The later interface shows that it was designed for mouse operation. In early versions snippets could be triggered only with the mouse, by right-clicking the recipient station. That remains possible.

Snippet
Hi QRZNAME, try sked 2m? Ur QRG?
try? FIRSTAP, pse lsn MYQRGSHORT
maybe we need AP, FIRSTAP SECONDAP
I am calling cq to ur dir, pse lsn to MYLOCATOR at MYQRG
pse ur qrg?
rrr, I move to your qrg nw, pse ant dir MYLOCATOR
Hrd you but many qrm here, pse agn
I turn my ant to you now
Sry, strong qrm by local station, may try MYQRG
Sry, in qso nw, pse qrx, I will meep you
Ur ant my dir MYLOCATOR nw?
QRZNAME pse ant dir MYLOCATORSHORT



Software-assisted optimisation of contest station operating practice using KST4Contest

IQ4KD and IU3OAR gave me a good reason to add keyboard access, using Ctrl plus a number to trigger the corresponding text. They convinced me. The reason is this miniature keyboard.

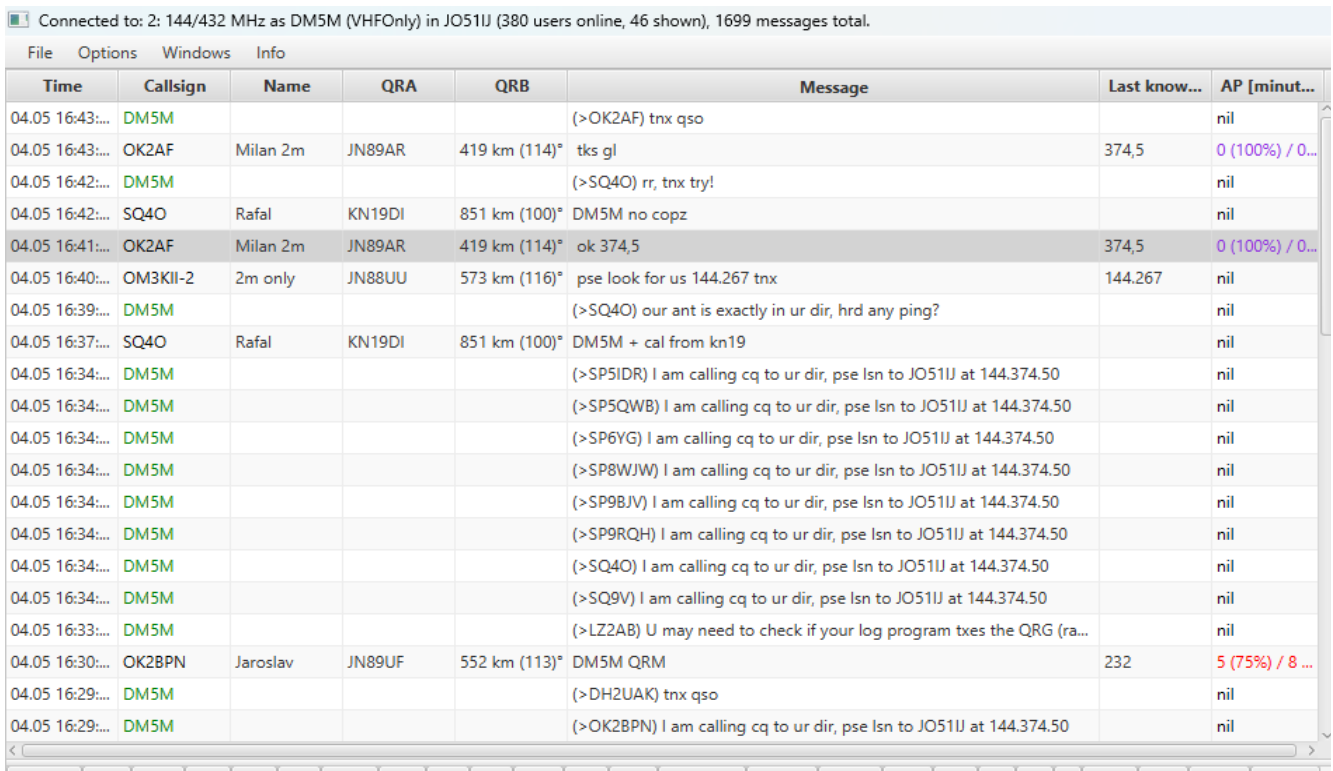
They, and I, place the prepared texts on this programmable miniature keyboard [13], available online for about EUR 20. It saves an enormous amount of time in the station workflow.[13]

I will now divide the screenshot into sections and explain each in terms of workflow and operating practice.

Software-assisted optimisation of contest station operating practice using KST4Contest

b) Private messages

Figure 3.1: Private-message section. This window shows incoming and sent private messages.



Connected to: 2: 144/432 MHz as DM5M (VHFOnly) in JO51IJ (380 users online, 46 shown), 1699 messages total.							
File	Options	Windows	Info				
Time	Callsign	Name	QRA	QRB	Message	Last know...	AP [minut...
04.05 16:43:...	DM5M				(>OK2AF) tnx qso		nil
04.05 16:43:...	OK2AF	Milan 2m	JN89AR	419 km (114)*	tkx gl	374,5	0 (100%) / 0...
04.05 16:42:...	DM5M				(>SQ4O) rr, tnx try!		nil
04.05 16:42:...	SQ4O	Rafal	KN19DI	851 km (100)*	DM5M no copz		nil
04.05 16:41:...	OK2AF	Milan 2m	JN89AR	419 km (114)*	ok 374,5	374,5	0 (100%) / 0...
04.05 16:40:...	OM3KII-2	2m only	JN88UU	573 km (116)*	pse look for us 144.267 tnx	144.267	nil
04.05 16:39:...	DM5M				(>SQ4O) our ant is exactly in ur dir, hrd any ping?		nil
04.05 16:37:...	SQ4O	Rafal	KN19DI	851 km (100)*	DM5M + cal from kn19		nil
04.05 16:34:...	DM5M				(>SP5IDR) I am calling cq to ur dir, pse lsn to JO51IJ at 144.374.50		nil
04.05 16:34:...	DM5M				(>SP5QWB) I am calling cq to ur dir, pse lsn to JO51IJ at 144.374.50		nil
04.05 16:34:...	DM5M				(>SP6YG) I am calling cq to ur dir, pse lsn to JO51IJ at 144.374.50		nil
04.05 16:34:...	DM5M				(>SP8WJW) I am calling cq to ur dir, pse lsn to JO51IJ at 144.374.50		nil
04.05 16:34:...	DM5M				(>SP9BJV) I am calling cq to ur dir, pse lsn to JO51IJ at 144.374.50		nil
04.05 16:34:...	DM5M				(>SP9RQH) I am calling cq to ur dir, pse lsn to JO51IJ at 144.374.50		nil
04.05 16:34:...	DM5M				(>SQ4O) I am calling cq to ur dir, pse lsn to JO51IJ at 144.374.50		nil
04.05 16:34:...	DM5M				(>SQ9V) I am calling cq to ur dir, pse lsn to JO51IJ at 144.374.50		nil
04.05 16:33:...	DM5M				(>LZ2AB) U may need to check if your log program txes the QRG (ra...		nil
04.05 16:30:...	OK2BPN	Jaroslav	JN89UF	552 km (113)*	DM5M QRM	232	5 (75%) / 8 ...
04.05 16:29:...	DM5M				(>DH2UAK) tnx qso		nil
04.05 16:29:...	DM5M				(>OK2BPN) I am calling cq to ur dir, pse lsn to JO51IJ at 144.374.50		nil

Figure 3.1: Private-message section3.1

Some data is visibly redundant. QRB and QRA appear both in the user information and the user list, and repeat in every row involving the same participant.

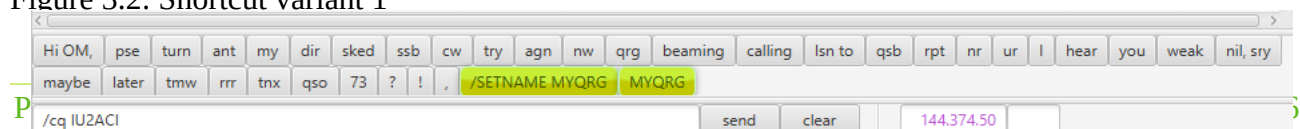
That is intentional.

When a sked or reply arrives, the station is often already in the calling process. Seeing all station data immediately helps select an antenna from the QRB or point it using the QRA. Searching a user list that sometimes has more than 400 entries would cost time. This is direct support for operating practice.

The table also shows the last known QRG detected by the message processor and the next two aircraft available for Aircraft Scatter. This information appears for every message for the reason just described.

c) Shortcuts, text input and QRG

Figure 3.2: Shortcut variant 1



Hi OM, pse turn ant my dir sked ssb cw try agn nw qrg beaming calling lsn to qsb rpt nr ur I hear you weak nil, sry
maybe later tmw rrr tnx qso 73 ? ! , /SETNAME MYQRG MYQRG
/cq IU2ACI send clear 144.374.50

Figure 3.2: Shortcut variant 13.2

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I saw shortcut buttons like these at DJ5AR's home. Andreas had convinced me that assembling text with the mouse was useful; I believe he used it to generate CW messages. I decided to implement the same idea.

The texts are freely configurable. In practice, complete macros on the shortcut bar are as useful as assembling phrases, as variant 2 shows. Writing standard text becomes very fast.

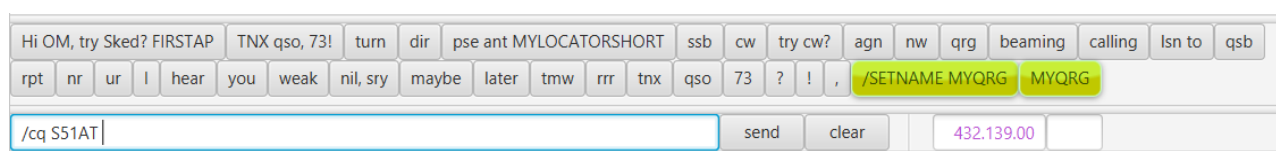


Figure 3.3: Shortcut variant 23.3

Figure 3.3: Shortcut variant 2

After a station or macro is clicked, the text input immediately receives focus so typing can begin. That also accelerates the process.

When the logger supports it, the QRG field receives its value from the logger, which polls the TRX. It can also be entered manually. The MYQRG snippet provides an immediate reply to another station asking for the QRG.

This saves a great deal of time because such questions occur constantly.

The empty field was reserved for the local bearing, MYQTF, to be supplied by PSTRotator. In January 2025 this was untested and therefore disabled. Once integrated, it could control QTF filters for the user list and CQ messages automatically, leaving still less work at the keyboard.

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d) User list

Show only QRB [km] <= 900.0

☒ Show only QTF: 90 deg +/- 50.0

N

NE

E

SE

S

SW

W

NW

Find...

Hide worked: wkd 144 Inactive stations

Hide un-QRV:

Callsign ▲	Name	QRA	QTF	QRG	Act	AP [minutes / pot...	worked		NOT QRV @
							144	wkdany	
OK6M	Martin	JN99CR	106.36°	432,182	1	0 (100%) / 0 (100%)			
OL4N	2m	JO60VR	107.78°	144242	0	2 (75%) / 3 (75%)			
OM3KDX-1	Radioclub	KN18DQ	105.49°		19	nil			
OM3KUK	RC DnV 144...	JN99BB	113.3°	432.204	3	nil			
OM3W	Club 2m	JN99CH	110.61°	396	0	nil			
OM3W-70	Club 70cm	JN99CH	110.61°	432.266 201	2	nil			
OM6DN	2x12ele 950...	JN99FI	109.43°	144.151	0	0 (100%) / 0 (100%)			
SN7L	Team 144.182	JO91QF	88.37°		10	0 (100%) / 0 (100%)			
SN9H	HS KRK	KO00TD	96.0°		30	5 (75%) / 13 (75%)			
SP5IDR	Andy 2m 70...	KO01VU	81.85°		30	2 (75%) / 11 (75%)			
SP5QWB	Bart	KO02NO	75.25°		30	0 (100%) / 1 (75%)			
SP6FU	Marcin 2m	JO80HN	100.3°		0	nil			
SP6YG	Mariusz 2m...	JO90CK	98.53°	144,159	10	nil			
SP8WJW	Piotr 2m	KN09TV	97.96°		11	4 (75%) / 22 (75%)			
SP9BJV	6/2/70	JO90KG	99.12°		30	0 (100%) / 1 (75%)			
SP9RQH	CHRIS 14 el...	JO90TC	99.41°		30	1 (75%) / 3 (75%)			
SQ4O	Rafal	KN19DI	100.93°		0	nil			
SQ9V	Theodor	JO90EB	102.37°	275	7	0 (100%) / 1 (75%)			

The user list includes the QRG among other information.

Callsigns in bold green have recently written towards my station within a configured opening angle, as SQ9V has here. Their antenna probably points towards me.

A per-band Worked flag marks stations already contacted in the contest. A NOT-QRV flag records stations known not to be QRV on a band. This prevents redundant requests and saves time. Many values act as filter markers.

Enabling the per-band Hide Worked filter removes stations already logged on that band. Callsigns can likewise be filtered by QRB and QTF, for example when calling CQ in a chosen direction while contacting matching stations.

Filtering saves considerable time and prevents mistakes when interpreting callsigns or QRAs.

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e) User-information window

Selecting a participant updates the user-information window. It includes a direction indicator, the green arrow, which starts tracking the station in AirScout when clicked.

The selected station can also be marked not QRV on a band and filtered later. Messages to and from it can be filtered, leaving a station-specific chat history that can help assess whether the QRG field is correct.

Messages of DM7D -> Filter: ☒ nothing ☐ pm to me ☐ pm to other ☐ public msgs

Time	Call TX	Call RX	Message
27.12 22:19:36	DG0VV	DM7D	rr 275
27.12 22:18:20	DM7D	ALL	(dg0vv) 144,275 ???
27.12 22:17:18	DG0VV	DM7D	rr2m 250
27.12 22:10:56	DM7D	ALL	(dg0vv) 144,250
27.12 22:10:29	DM7D	ALL	(dg0vv) auf 2 m
27.12 22:05:36	DG0VV	DM7D	ronald wo bist du ich rufe dich

QTF:42.89 deg ☒ Show path in AS ☐ tag not qrv 144

QRB: 448.96 km ☐ Lookup on qrz.com ☐ tag not qrv 432

Last activity: 27.12 22:18:21 ☐ Lookup on qrzcq.com ☐ tag not qrv 23cm

(24 min ago) ☐ tag not qrv all

f) CQ window

Time	Callsign	Name	Message	Last QRG
04.05 16:44:...	DF2FQ	Holger	OK6M I'm listening	432.295
04.05 16:44:...	SQ4O	Rafal	9a1ct + trz	
04.05 16:43:...	DM5M		~"~"~"~"~"~" CQ CQ CQ de DM5M, pse call us at 144.374.50 ~"~"~"~"~"~"	
04.05 16:43:...	YT5W	RK Pozarevac	cq 155 beam 9A S5 IZ	155
04.05 16:43:...	I5WBE	Enrico	IZ3QFG No dopo spengo passami solo il progressivo lentamente	
04.05 16:43:...	OM6DN	2x12ele 950...	>>> anyone for sked<<<<	144.151
04.05 16:42:...	OE6PPF	Peter only 2m	(OM2Y) pse beam to me	193
04.05 16:42:...	I4GHG	Rino 144.390	JN63dt - 1200 mt asl!!!!'m not taking part in the contest, anyway welcome USB D...	144.392
04.05 16:42:...	LZ2AB	144.010.00	~"~"~"~"~"~" CQ CQ CQ de LZ2AB, pse call us at 144.010.00 ~"~"~"~"~"~"	144.010.00
04.05 16:42:...	YU7ACO	2m Radio cl...	IK3TPP pse sked on 144.200	144.200
04.05 16:42:...	YU7ACO	2m Radio cl...	IK2PTR pse sked on 144.200	144.200
04.05 16:42:...	SQ4O	Rafal	DM5M no copz	
04.05 16:42:...	I5WBE	Enrico	IZ3QFG Ok per il loc.	
04.05 16:42:...	DM7D	Ronald	(om3KII) we are still on 267	267

The CQ window shows messages addressed to everyone. It is the proper place for automatic text beacons, which KST4Contest can send at selectable intervals. QRG details are captured and updated throughout the program.

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g) Other stations' QSOs

I almost never keep this window open. It shows QSOs between other stations, but often contains too much traffic to monitor continuously.

The text is still a data source, for example for probable calling direction and QRG information.

Some users proposed a QSO spy like the one in KST2Me so that nearby stations could be monitored deliberately. That is what some operators try to do in this window. I planned a QSO spy for the next version to make selected monitoring convenient.

Time	Call TX	Last QRG TX	wkd TX?	Call RX	Last QRG RX	wkd RX?	Name	
02.07 12:49:...	OK1KQH	144.398		DF0YY	432.179	X	Radioclub (...)	ok tnx 73
02.07 12:49:...	DK0A	144.237		F4FAQ	432	X	Club (1140...	pse sked 144.237
02.07 12:49:...	OK2O	144.317		DL8UWE		X	2m 317	GA Uwe, PSE sked on 144.317 OK?
02.07 12:49:...	HG6N	235		OK2KOJ	235		Tibor	OK, call me please.
02.07 12:49:...	YT3N	360		IK4GNG	267		Nikola 6/2/...	ok
02.07 12:49:...	SP2WPY			SQ9V	432.340		Slawomir	rrr
02.07 12:49:...	S59ABC	144193		YO4FYQ	432.211	X	Radio klub	pse sked my freq 144193
02.07 12:49:...	OK6RA	144.390		YT5DEY	250	X	Vaclav 2m ...	Hello. 144.390 pls?
02.07 12:49:...	IK3XTT	235		9A0V-7	432237		Only 432	i copy oly number
02.07 12:49:...	G3XDY	144.210		MM0GPZ		X	John	Freq on 2m Gordon?
02.07 12:49:...	OM6DN	144.255		UA2FZ	432		2x12ele 950...	tnx
02.07 12:49:...	IZ3NOC/5	432.198		DL8UWE			Fede 432 o...	Hi Uwe ?
02.07 12:49:...	YO2GL	082		DK0NA(left)			Karl	ok 082 cw pse kn05 timisoara
02.07 12:49:...	SQ9V	432.340		SP2WPY			Theodor	Ssb
02.07 12:49:...	OK1CRM	144.162		DF6WE		X	Palo Only 2m	tnx
02.07 12:49:...	IZ1AZA	346		S50TA			Alex 1400m...	sorry no 70
02.07 12:49:...	OK2AF	390		DF2AJ		X	Milan 2m	pse 2m sked
02.07 12:49:...	S50TA			OE3A	160		Andrej	pse 70 cm qrg
02.07 12:49:...	IQ3LX	144.288		9A1AAY	144288		QRG 144.290	tnx for qso
02.07 12:49:...	IK4GNG	267		YT3N	360	X	Gibo only 1...	267 ok ?
02.07 12:49:...	SP2WPY			SQ9V	432.340		Slawomir	rrr, CW?
02.07 12:49:...	SP5U	057		OL2J	144.057		Mirek	Only possible via AP
02.07 12:49:...	LZ6R	432.210		OK6M	432.360		Svet LZ2SQ	ok
02.07 12:49:...	OK6RA	144.390		YT3N	360	X	Vaclav 2m ...	Hello. 144.390 pls?
02.07 12:49:...	PA0O	395		OM3KII	432.310		Jaap 23 70	395
02.07 12:49:...	YP2DX	432280		IK6SIO	383		Club	GE Frank, pse sked on 70cm
02.07 12:49:...	YU7KB	144365		I0YLI	387		Bane	pse 144365
02.07 12:49:...	9A6K	244 239		DJ2DA	432.240		70-23-13-9-3	pse 244 now local qrm 239...
02.07 12:49:...	HA6W	144.212		DB4MM		X	Team 2m	We are in ur dir now on 144.212, AP is now!
02.07 12:49:...	OK1GK	432.395		S57LM	395	X	2m+70cm ...	tnx for try
02.07 12:49:...	OK6RA	144.390		YP2DX	432280	X	Vaclav 2m ...	Hello. 144.390 pls?
02.07 12:49:...	OM2Y	144.170		IK2RLN		X	Jan	GA, pse try qso on 144.170
02.07 12:49:...	YP2DX	432280		9A5AFF	256 258		Club	GE Dean, pse sked on 70cm

Outside central Europe, for example in SM or LZ, this makes sense and can produce more QSOs. Following the activity of nearby stations allows their trail to be picked up and their counterpart to be worked when an opportunity arises. Alternatively, a DX station can be watched and contacted when it works into one's own locator area.

4) Less visible assistance from KST4Contest

So far I have shown how visible data was intended to support the contest workflow. There are subtler functions too. In my view, the most important is the treatment of misaddressed messages.

a) Misaddressed messages

There are two message categories in the chat channel:

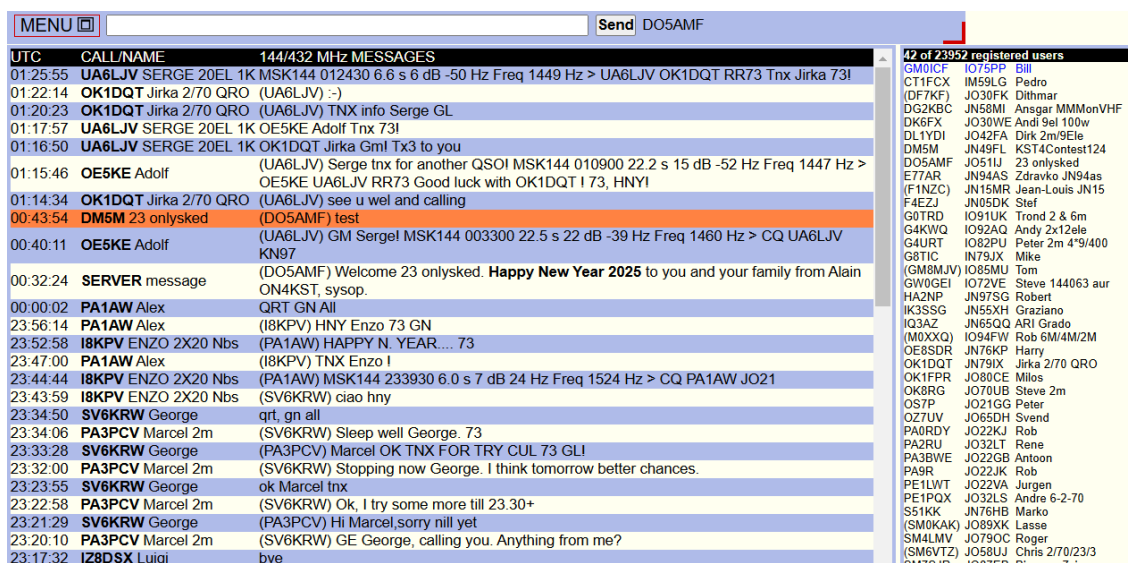
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- **messages to everyone, or public messages;**

Every user can read every message, but public messages have no explicit recipient. An example input is “GM dr OMs, gl in contest”.

- **messages with a specified recipient.**

Addressed messages trigger a highlight and an audible warning. An example input is “/CQ DO5AMF hi marc, gl in contest”.



The screenshot shows the ON4KST web client interface. The top bar includes a 'MENU' button and a 'Send' field with 'DO5AMF' entered. The main area is divided into two columns. The left column, titled '144/432 MHz MESSAGES', displays a list of messages with columns for UTC, CALL/NAME, and the message text. The message '00:43:54 DM5M 23 onlksked (DO5AMF) test' is highlighted in orange. The right column, titled '42 of 23952 registered users', lists various call signs and names, including CT1FCX, IM59LG, Pedro, (DF7KF), JO30FK, Dithmar, DG2KBC, JN58MI, Ansgar MMMonVHF, DK6FX, JO30WE, Andi 9el 100w, DL1YDI, JO42FA, Dirk 2m/9Ele, DM5M, JN49FL, KST4Contest124, DO5AMF, JO51U, 23 onlksked, E77AR, JN94AS, Zdravko JN94as, (F1NZC), JN15MR, Jean-Louis JN15, F4EZJ, JN05DK, Stef, G0TRD, IO91UK, Trond 2 & 6m, G4KWQ, IO92AQ, Andy 2x12ele, G4URT, IO82PU, Peter 2m 4*9/400, G8TIC, IN79JX, Mike, (GM8MJV), IO85MU, Tom, GW0GEI, IO72VE, Steve 144063 aur, HA2NP, JN97SG, Robert, IK3SSG, JN55XH, Graziano, IQ3AZ, JN65QQ, ARI, Grado, (M0XXQ), IO94FW, Rob 6M/4M/2M, OE8SDR, JN76KP, Harry, OK1DQT, JN79IX, Jirka 2/70 QRO, OK1FPR, JO80CE, Milos, OK8RG, JO70UB, Steve 2m, OS7P, JO21GG, Peter, OZ7UV, JO65DH, Svend, PA0RDY, JO22KJ, Rob, PA2RU, JO32LT, Rene, PA3BWE, JO22GB, Antoon, PA9R, JO22JK, Rob, PE1LWT, JO22VA, Jurgan, PE1PQX, JO32LS, Andre 6-2-70, S51KK, JN76HB, Marko, (SM0KAK), JO89XK, Lasse, SM4LMV, JO79OC, Roger, (SM6VTZ), JO58UJ, Chris 2/70/2/3, and a 'bye' message at the bottom.

The screenshot shows “/CQ DO5AMF test” from DM5M highlighted in the ON4KST web client. Imagine how quickly it disappears when four messages arrive each second.

That alone is challenging. If the sender omits “/CQ” or writes “(DO5AMF) Test” without knowing the syntax, the web client produces no highlight. During heavy activity, the message is very unlikely to be read.

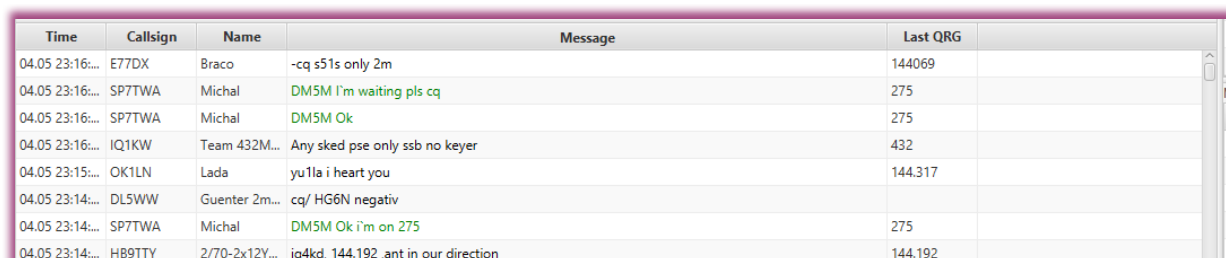
1) Countermeasure

If the missed message is a reply to one's sked request or a note from a station already listening that the antenna should be turned, the intended recipient probably loses the chance of a quick QSO. At the busiest time it may cost the contact; at best it wastes time and work.

I therefore place messages containing my callsign in the private-message window even when they were accidentally sent to everyone or addressed elsewhere, and add an audible indication. The user need not notice the addressing mistake.

Two screenshots from a previous contest illustrate this:

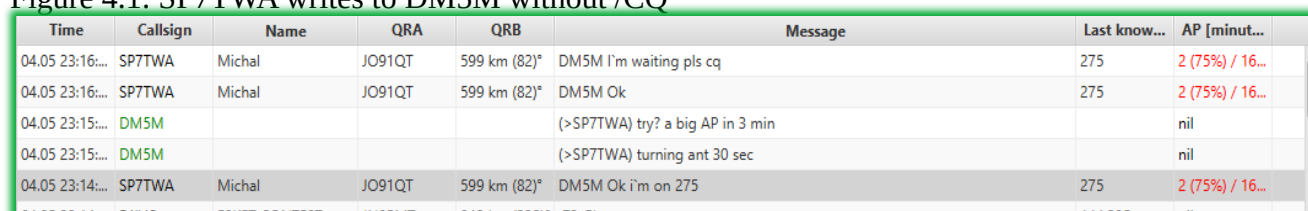
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Time	Callsign	Name	Message	Last QRG
04.05 23:16:...	E77DX	Braco	-cq s51s only 2m	144069
04.05 23:16:...	SP7TWA	Michal	DM5M I'm waiting pls cq	275
04.05 23:16:...	SP7TWA	Michal	DM5M Ok	275
04.05 23:16:...	IQ1KW	Team 432M...	Any sked pse only ssb no keyer	432
04.05 23:15:...	OK1LN	Lada	yu1la i heart you	144.317
04.05 23:14:...	DL5WW	Guenter 2m...	cq/ HG6N negativ	
04.05 23:14:...	SP7TWA	Michal	DM5M Ok i'm on 275	275
04.05 23:14:...	HB9TTY	2/70-2x12V...	iq4kd, 144.192, ant in our direction	144.192

Figure 4.1: SP7TWA writes to DM5M without /CQ4.1

Figure 4.1: SP7TWA writes to DM5M without /CQ



Time	Callsign	Name	QRA	QRB	Message	Last know...	AP (minut...
04.05 23:16:...	SP7TWA	Michal	JO91QT	599 km (82)*	DM5M I'm waiting pls cq	275	2 (75%) / 16...
04.05 23:16:...	SP7TWA	Michal	JO91QT	599 km (82)*	DM5M Ok	275	2 (75%) / 16...
04.05 23:15:...	DM5M				(>SP7TWA) try? a big AP in 3 min		nil
04.05 23:15:...	DM5M				(>SP7TWA) turning ant 30 sec		nil
04.05 23:14:...	SP7TWA	Michal	JO91QT	599 km (82)*	DM5M Ok i'm on 275	275	2 (75%) / 16...

Figure 4.2: Messages from SP7TWA arrive in the private-message window4.2

Figure 4.2: Messages from SP7TWA arrive in the private-message window. The user does not notice the addressing mistake.

2) A note

Strictly speaking, operating practice in the chat is not amateur-radio operating practice because it is Internet rather than radio. Language rules and conventions still apply:

- **address messages with /CQ <CallSign>;**
- abbreviations from CW are customary;
- the operating language is English;
- the operator name in the chat often contains band information.

b) Extracting more from the chat

Computer scientists have a tendency to collect and analyse available data. The chat client should be no exception. Looking at the data exchanged raises the question of what else can be done with it. I have already mentioned the first example.

1) Frequency extraction

The phrase sounds unwieldy, but I had no better one. Anything that looks like a frequency is written to the frequency field of the station that sent the text. There is no convention for writing a QRG in the chat, which is a particular challenge. Some direct examples follow:

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- “pse sked on 144.240”
- “Pse sked 144,270”
- “PSE QSO 144349.5”
- “pse look for me on 432264”
- “PSE SKED OM3KHU .224”
- “pse try 162 ?”

The MHz separator may be a point, comma or space, or absent entirely. Some values have three digits while others use seven or eight, even on 144 MHz. I developed several regular expressions.

One example from my working notes matches QRG forms such as 144223, 144223.xx, 144,223,xx and 144.223.xx: “((([0-9]{3,4}[\.,|]?[0-9]{3})([\.,|]\d{1,2})?)”

Callsigns with prefixes required a more complicated expression: (([a-zA-Z]{1,2}[\d{1}]?V)?(\d{1}[a-zA-Z][\d{1}][a-zA-Z]{1,3})(Vp)(\d{1})?)([a-zA-Z0-9]{1,2}[\d{1}]?V)?([a-zA-Z]{1,2}(\d{1,3}[a-zA-Z]{1,4})))((Vp)(\d{1})?)”

Regular expressions apply a defined format very quickly to large data sets. An excellent test environment is listed in source [14].[14]

I learned only in August 2024 that others had already addressed this problem, after my own solution was adequate. I happened to notice potential QRG values highlighted in the wtKST chat review. I had not known either that highlighting existed or that there was a chat review at all.

For three-digit values, interpreting the band remains the recipient's responsibility. I left it that way because an uninformed message processor has limited means.

G1YFG and I were working on a proposed solution: a shared open protocol for exchanging frequencies and QRV information through a jointly used, automatically maintained web service. This could also support the message processor.

2) Highlighting a sked directed to me

Another cumbersome phrase that explains little. A diagram and short description therefore come first.

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- **Message from F5FEN to DM5M:**

DO5AMF in JN49 and
DO5AMF/P see the highlight

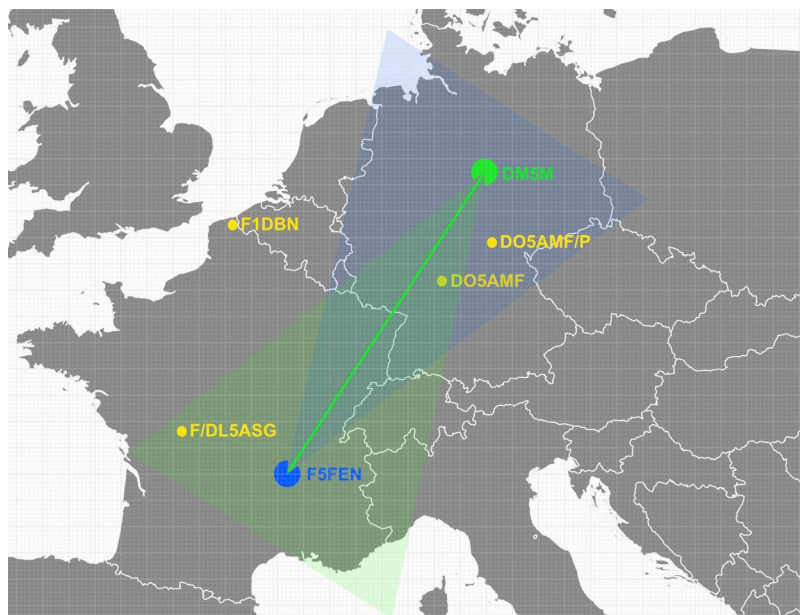
F/DL5ASG does not see it

F1DBN sees no highlight

- **Message from DM5M to F5FEN:**

DO5AMF in JN49 sees the
highlight

DO5AMF/P and F/DL5ASG do
not; F1DBN sees no highlight



I assume that a station contacting another has turned its antenna in that direction. This matches many samples. The station is then transmitting towards me, and I can work it almost incidentally. Because the message processor has usually captured the QRG, I can simply try. DO5AMF, including at his portable location, may benefit from DM5M's sked with F5FEN and line up for F5FEN directly.

The success rate of this mechanism, measured by tally marks by others and me, was about 60%. The stations might have been worked another way, of course. The aim is to use favourable opportunities quickly. The function supports operating practice because the operator need not draw the conclusion manually from the chat history.

People have repeatedly noticed that DM5M often appears on a QRG shortly after another station's sked. This mechanism is responsible.

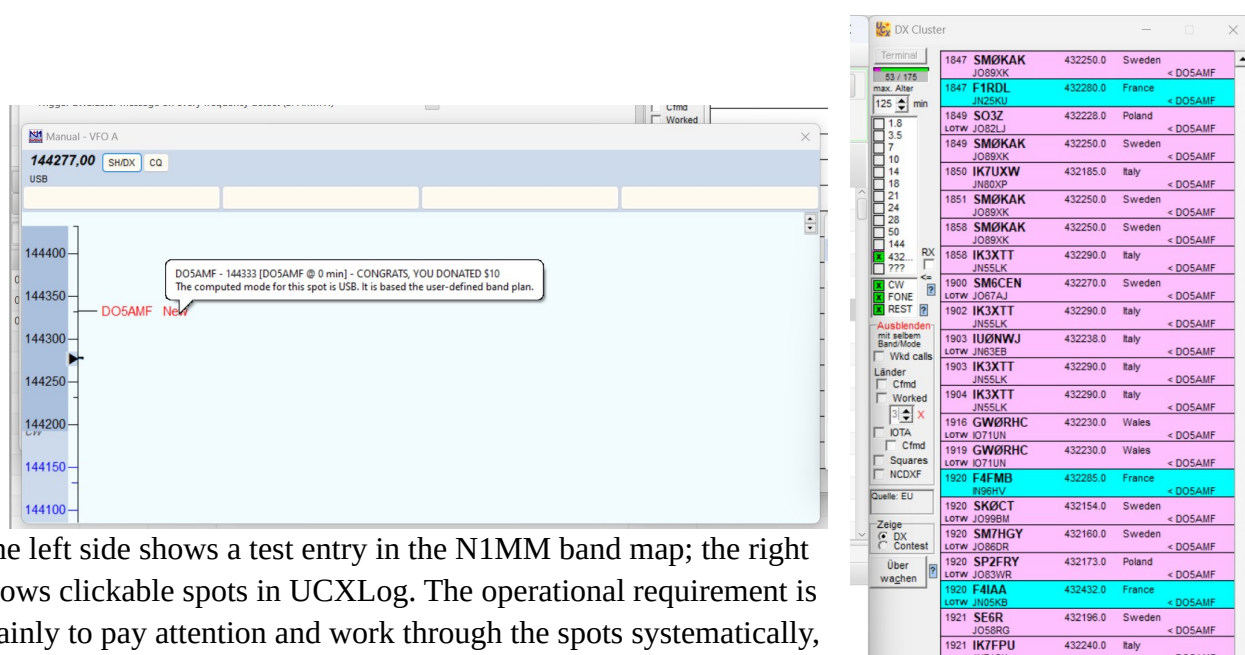
The opening angles are configurable and should be reasonably realistic. At the time, path propagation was not considered at all. It might be possible to use the AirScout interface, or to integrate and evaluate a path-elevation diagram, perhaps with the Google Maps API [15].
[15]

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c) DX Cluster server

A cluster output was the logical next step because highlighting in the user list did not fit the station workflow well. OM0AAO supplied the idea. A cluster spot must, incidentally, contain a correctly formatted frequency.

Exact QRGs need only format normalisation. For three-digit values, the operator can configure an MHz prefix for KST4Contest to add, which occasionally produces a spot on the wrong band. As mentioned, a better solution was being considered. Cluster spots complement the user-list highlight and, in UCXLog for example, a double-click can tune the TRX. The spots can be worked systematically with little tuning when the transmitting station described its QRG precisely.



The left side shows a test entry in the N1MM band map; the right shows clickable spots in UCXLog. The operational requirement is mainly to pay attention and work through the spots systematically, preferably as they appear.

A second operator at a separate interlocked TRX, or a FlexRadio with a second slice in SO2R operation, creates effective ways to collect opportunistic QSOs.

I did not implement a conventional cluster because it promised little additional value, although it would be possible.

5) Interfaces

Here “interface” means the software-development kind, not very different in principle from a connector on a radio: an interaction point between software components or systems.

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Two programs agree unambiguously on a communication mechanism: a protocol such as TCP or UDP over IP, source and destination addresses, and a destination port. Source addresses and ports do not matter to the UCXLog and AirScout interfaces.

For a limited broadcast on the local network, the destination is 255.255.255.255. Routers do not forward such packets through the gateway. AirScout and UCXLog use this method; the sender neither knows nor checks whether the receiver obtained a packet.

The message format must also be defined. In these cases it is plain-text CSV or XML.

a) Logger interface using ContactInfo and RadioInfo packets

As mentioned earlier, obtaining log data works fundamentally differently from Win-Test with wtKST, where the chat client communicates directly with the contest logger through a defined API. UCXLog and several other programs provide no such API.

The simple log file did not provide enough information, so I considered how to obtain what was needed. Simulating a UCXLog client on the network would have been possible but laborious, and it would have interfered with the logger's transaction logic. That should be avoided to protect log consistency.

It would also have created vendor lock-in to one logger. I wrote to DL7UCX. He explained the database structure but understandably preferred that I not access it directly. Instead, he suggested listening to the ContactInfo and RadioInfo broadcasts through which UCXLog reports what it has stored.

I began the next day, and two days later the interface was working. It proved an excellent idea. Research showed that many loggers transmit these packets, making KST4Contest compatible without changes with QARTest, N1MM+, DXLog, Ham Radio Deluxe, Log4OM and UCXLog. VQLog and Win-Test planned to add support, with Win-Test already transmitting RadioInfo. There may be more; I do not keep the list current.

1) ContactInfo packet

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The parameters in a ContactInfo packet are shown here. Readers may imagine what else could be done with them. At the time I was implementing upload to the SLOVHF scoreboard following requests from KST4Contest users. This was comparatively simple because KST4Contest did not have to calculate points, only add them.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <contactinfo>
3   <app>N1MM</app>
4   <contestname>CWOPS</contestname>
5   <contestnr>73</contestnr>
6   <timestamp>2020-01-17 16:43:38</timestamp>
7   <mycall>W2XYZ</mycall>
8   <band>3.5</band>
9   <rxfreq>352519</rxfreq>
10  <txfreq>352519</txfreq>
11  <operator></operator>
12  <mode>CW</mode>
13  <call>W1AW</call>
14  <countryprefix>K</countryprefix>
15  <wprefix>W1</wprefix>
16  <stationprefix>W2XYZ</stationprefix>
17  <continent>NA</continent>
18  <snt>599</snt>
19  <sntnr>5</sntnr>
20  <rcv>599</rcv>
21  <rcvnr>0</rcvnr>
22  <gridsquare></gridsquare>
23  <exchangel></exchangel>
24  <section></section>
25  <comment></comment>
26  <qth></qth>
27  <name></name>
28  <power></power>
29  <misctext></misctext>
30  <zone>0</zone>
31  <prec></prec>
32  <ck>0</ck>
33  <ismultiplier1>1</ismultiplier1>
34  <ismultiplier2>0</ismultiplier2>
35  <ismultiplier3>0</ismultiplier3>
36  <points>1</points>
37  <radionr>1</radionr>
38  <run1run2>1</run1run2>
39  <RoverLocation></RoverLocation>
40  <RadioInterfaced>1</RadioInterfaced>
41  <NetworkedCompNr>0</NetworkedCompNr>
42  <IsOriginal>False</IsOriginal>
43  <NetBiosName></NetBiosName>
44  <IsRunQSO>0</IsRunQSO>
45  <StationName>CONTEST-PC</StationName>
46  <ID>f9ffac4fcd3e479ca86e137df1338531</ID>
47  <IsClaimedQso>1</IsClaimedQso>
48  <oldtimestamp>2020-01-17 16:43:38</oldtimestamp>
49  <oldcall>W1AW</oldcall>
50  <SentExchange>XYZ NY</SentExchange>
51 </contactinfo>
```

2) RadioInfo packet

At the time, the RadioInfo packet supplied only the QRG. Its data also suggests further possibilities, such as the selected rotor and direction.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <RadioInfo>
3   <app>N1MM</app>
4   <StationName>CW-80m</StationName>
5   <RadioNr>1</RadioNr>
6   <Freq>352211</Freq>
7   <TXFreq>352211</TXFreq>
8   <Mode>CW</Mode>
9   <OpCall>W1ABC</OpCall>
10  <IsRunning>False</IsRunning>
11  <FocusEntry>204626</FocusEntry>
12  <EntryWindowHwnd>275678</EntryWindowHwnd>
13  <Antenna>8</Antenna>
14  <Rotors></Rotors>
15  <FocusRadioNr>1</FocusRadioNr>
16  <IsStereo>False</IsStereo>
17  <IsSplit>False</IsSplit>
18  <ActiveRadioNr>1</ActiveRadioNr>
19  <IsTransmitting>False</IsTransmitting>
20  <FunctionKeyCaption></FunctionKeyCaption>
21  <RadioName></RadioName>
22  <AuxAntSelected>-1</AuxAntSelected>
23  <AuxAntSelectedName></AuxAntSelectedName>
24  <IsConnected></IsConnected>
25 </RadioInfo>
```

These and other packets are documented in detail in the N1MM+ web documentation [16]. [16]

b) Interface to AirScout

I reproduced the AirScout interface, much as I had the basic ON4KST protocol, after analysing communication between wtKST and AirScout with Wireshark. In retrospect, asking the author would have been easier, faster and wiser. After contacting DL8AAU in August of the previous year,

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I am convinced he would have provided the information without hesitation. The idea simply did not occur to me. Fortunately, reverse engineering it was not an enormous effort.

Requests from the chat client use UDP, and the protocol itself contains control information defining sender and recipient. The example packet shows a KST4Contest request to AirScout for aircraft from DM5M to OK1MZM, with comments from my working notes. One corresponding AirScout reply follows.

```
243 ASSETPATH: "KST" "AS" 1440000,DM5M,JO51IJ,OK1MZM,JN89IW
244           ^^^own
245           ^^^assrvr
246           ^^^^^^band
247           ^^^^^owncall
248           ^^^^^^ownloc
249           ^^^^^^othercall
250           ^^^^^^otherloc
```

```
252 ASNEAREST: "AS" "KST" "2023-03-09 21:34:10Z,DM5M,JO51IJ,OK1MZM,JN89IW,
253 3, DHK4983,M, 62, 50, 5, TAY4367,M, 4, 100, 0, MNB9345,H, 8, 100, 0"
```

The next n aircraft are returned in this order: identifier, aircraft size where 0 is unavailable, 1 light, 2 heavy and 3 super-heavy, reflection potential, and time until arrival at the reflection point.

There are further packets that I omit because the paper is already very long.

c) Interface to the cluster

The cluster interface is an exception because it uses neither UDP nor CSV or XML. There is no strict syntax. A TCP connection using Telnet is established between client and server, and text resembling the example is exchanged.

```
DX de D05AMF: 144333.0 D05AMF CONGRATS, YOU DONATED $100 2315Z
```

While implementing this function, I noticed that logger developers interpret cluster information differently. DL7UCX handles it much as I would: presumably all whitespace in the incoming string is replaced by a separator and the string is split at that separator, allowing the data to be parsed cleanly. From the fifth field to the timestamp, the remaining fields contain the comment. N1MM+ took another route and required a defined number of spaces between values. Again, asking the author would probably have been sensible. Once more, however, the urge to experiment took over, and I kept trying until it worked. The night was young.

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d) Interface to the rotor

For rotor control I chose PSTRotator. Communication again uses XML over UDP.

An example packet looks like this:

```
1 <PST>
2   <AZIMUTH>85</AZIMUTH>
3 </PST>
```

I implemented the interface but disabled it. At the time I had no suitable rotor for testing, and I do not publish untested functions. Buying one merely for experiments did not seem attractive, while DM5M had other planned purchases that appeared more useful.

6) Integrating the software into contest operation

As described above, contesting today differs greatly from the way I experienced it at the beginning of my contesting life. I am a computer scientist and therefore have an intrinsic urge to play with technology. At the same time, I understand the views of testers who were and are successful without a chat and do not want radio operation to depend on a medium other than radio.

The chat is something one has to embrace. That is a process inherent in our hobby. For testers, a 24-hour event is the ideal opportunity to test their own resilience and that of the equipment.

It is simply a fact that a contest is more enjoyable when more QSOs take place. It resembles taking an alternative route around a motorway traffic jam: one reaches the destination either way, but it is nicer to keep moving. The result is immediate.

My approach is therefore to create acceptance through usability. The aim is to support the operator who uses the chat and, where appropriate, take over parts of the operating workload.

For newcomers to the chat in particular, the client must attract enough attention without becoming annoying.

I therefore try to avoid unnecessary screen movement, automate or shorten tasks, and use information that the operator never sees - in other words, abstract away from too much information.

The following are the lessons learned from our experience at DM5M:

- **Use the chat if you do not already do so. It helps make the most of the available equipment.**

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- Users must understand that repeated requests can make sense and do not oblige anyone to change QRG when the timing is unsuitable.
- New chat users need to adapt their habits. The client should remain in view and be considered as part of the operating process.
- Automatic beacon text should be disabled while moving quietly across QRGs.
- CQ messages should state the calling direction as well as the QRG where possible.
- Unworked stations in the current calling direction should be contacted, at the latest, during the final two thirds of the contest.
- Stations that remain QRV only briefly should be worked as a priority when they appear.
- Cluster spots should be worked regularly and preferably in parallel.
- The station's own frequency information should always be precise to avoid questions and mistakes.

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7) To-do list

This section copies my working notes. The titles are in English because I develop everything in English, as many software developers do.

- QSO Spy
 - planned for version 1.25: deliberately monitor the traffic of selected callsigns
- Warning: short QRV-time station may be worked using historical QRV-span data
 - Warn when a station is QRV that is usually available only briefly
- Mass mailings
 - Send identical text simultaneously to many stations in one direction
- Unworked grid-square warning for NAC
 - Highlight stations in locator squares not yet worked
- QRB in user list, completed
- Distance filter for the Aircraft Scatter wish list, bug
 - At present every station in the user list is copied to the Aircraft Scatter watch list
- Dark theme and configurable GUI settings
 - SM6VTZ wrote: “bring more color to the people”
- Line wrapping
 - Wrap long text and make it possible to copy
- “Auto 73” including checksum, for discussion
 - The idea was to send an automatic 73 when a station was logged, possibly with a checksum derived from the hexadecimal interpretation of report and locator. The counterpart could verify that the data had been received correctly without deriving it from the message. Would that be legal?
- “Back to CQ QRG” spot after QSY
 - After QSY, create an internal self-spot for the former CQ QRG so that the TRX can return to it without being touched
- Protocol and web service for exchanging chat users' QRG and QRV-band information
 - Work with G1YFG to develop an exchange protocol for QRG data

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8) Facts and figures

a) Development time and effort

The chat client had been under development since mid-2022, usually after the children were in bed and for no more than two or three hours a day. Without exaggeration, hundreds of hours went into it. Tests on Tuesdays during NAC and during IARU contests are not included. Producing the user manual sounds trivial, yet even the first version required 22 net hours.

b) Thanks

I owe the fact that the chat client ever left the development environment to Konrad Neitzel, DC9DJ. I only learned that he was a radio amateur after approaching him, as a professional software developer from the Java forum, with my deployment problem. I am very grateful. Users also test the software before publication and report their findings; their callsigns appear throughout the document. Many thanks to them.

c) Size of the project

The complete project then contained about 45,000 lines of code. The user community was steadily growing and, according to update requests, had been in three figures for some time. I regularly received requests, ideas and feedback, for which I was also grateful.

d) It will not make you rich

The client is primarily developed for DM5M and for my own use, so that is not a problem. But since people say one should not talk about money, and I am a rebel, I will do so anyway. In January 2025 the grand total of contributions was:

- **EUR 20**

Well, there we are. If you use the client, like it or want to support further development, the application and user manual explain how. Contributions of code are welcome too.

The project and its source code were publicly available on GitHub.

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9) Download

At the time of the presentation, the client and user manual were available from:

- <https://do5amf.funkerportal.de/>[17]

They could be downloaded there. After registering a domain at the end of 2024 for the planned QRG and QRV-data web service, I also intended to move the downloads to:

- <https://hamradioonline.de/>[18]

I expected both addresses to remain available for some time.

Contact for questions about the client: praktimarc+kst4contest@gmail.com

73, and see you in the contest. DO5AMF, Marc at DM5M

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